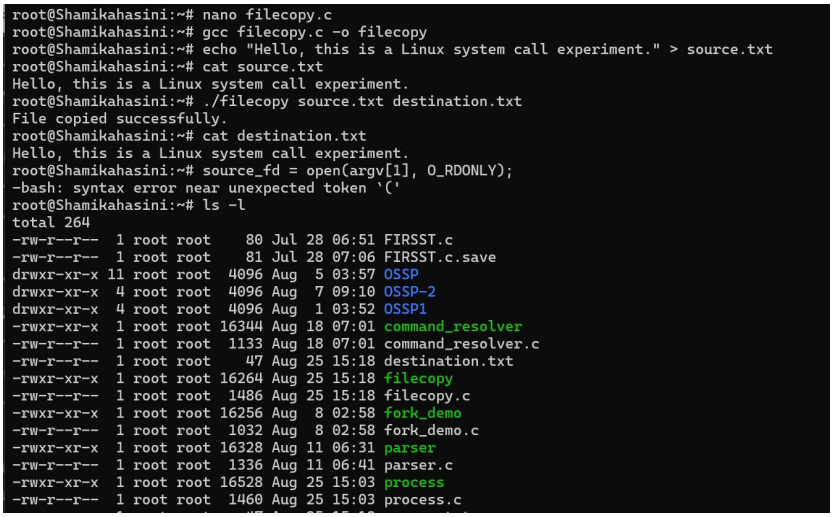
Practical-week-2

Task-1

Develop a C program that uses the system calls open(), read(), write(), and closeo to copy the contents of one file to 3 another. Explain how control transitions between user Space and kernel space during execution.

```
 GNU nano 7.2                                          filecopy.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source_fd, destination_fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    // Check command-line arguments
    if (argc != 3)
    {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    // Open source file for reading
    source_fd = open(argv[1], O_RDONLY);

    if (source_fd == -1)
    {
        perror("Error opening source file");
```

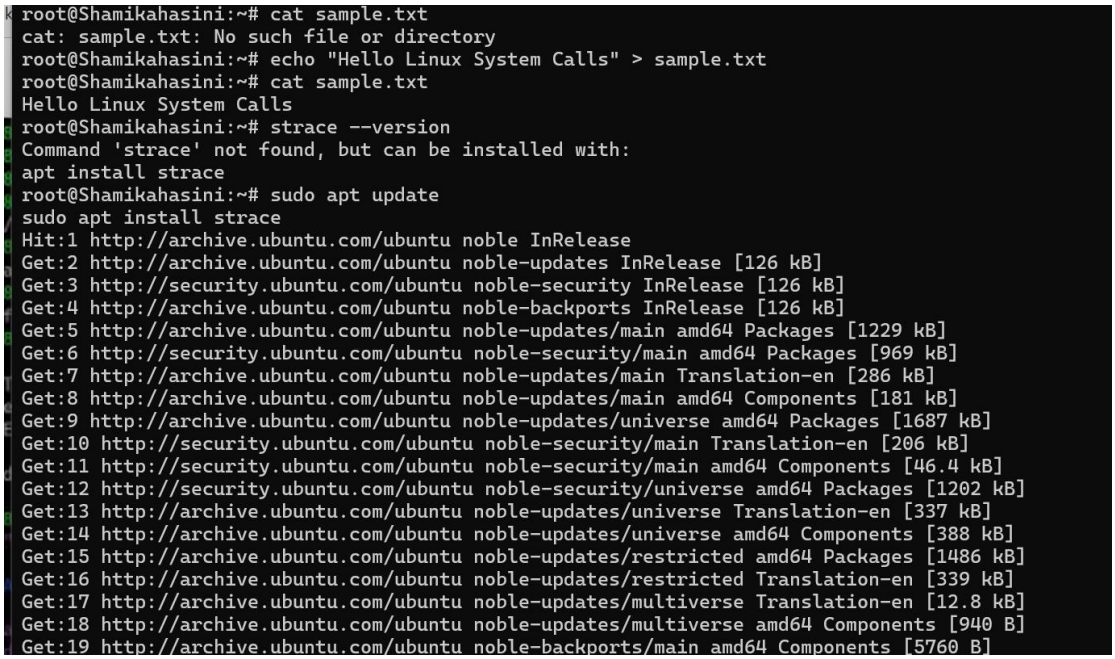
```
    destination_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination_fd == -1)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(destination_fd, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            close(source_fd);
            close(destination_fd);
            return 1;
        }
    }
```



Observation: The program successfully copied the contents of source.txt to destination.txt using the open(), read(), write(), and close() system calls. The contents of both files were verified to be identical. During execution, the user-space program requested file operations from the Linux kernel through system calls, and the kernel performed the required file operations before returning control to the user-space program

Task-2

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system 3 calls and identify the kernel services involved



Observation:

The strace utility successfully traced the system calls generated by the cat sample.txt command. The trace showed openat() being used to open sample.txt, followed by file and terminal operations using splice(), pipe(), and fcntl(). The close() system call was used to close the file descriptors, and exit_group(0) indicated successful program termination. The observed system calls demonstrate how the cat command interacts with Linux kernel services to access and display file contents.